

DTSA 5840

Data Science Applied Capstone Project

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Data Scientist (85.1 / 100)	71.6 / 100
SPECIALIZATION Applied Capstone		

72

CURRENT AVG SCORE
across 6 job roles

94

POTENTIAL SCORE
with recommended additions

+28

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

The Data Science Applied Capstone Project course will allow you to apply the knowledge and skills from the MS-DS degree to a real-world data set provided by industry collaborators and connections. This project will allow you to work independently on a data set that will test your skills in acquiring, cleaning, modeling data, and analyzing a dataset using data mining and machine learning techniques. By the end of this course, you will have a project that you can add to your data science portfolio to show off to employers and demonstrate your data science expertise. It is strongly recommended that you take the Data Science Applied Capstone Project course as one of your final courses in the program as you will work with real-world data sets that will use MS-DS core concepts. In order to be successful in this course, only students who have completed the following specializations should register for the Capstone Project: Data Science Foundations: Data Structures and Algorithms, Data Science Foundations: Statistical Inference, Data Mining Foundations and Practice and Machine Learning.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	85.1	100.0	+14.9	LOW
AI Engineer	81.1	100.0	+18.9	MEDIUM
Machine Learning Engineer	80.6	100.0	+19.4	MEDIUM
ML Engineer	72.3	98.4	+26.1	MEDIUM
Applied Scientist	64.5	92.5	+28.0	MEDIUM
MLOps Engineer	46.0	73.0	+27.0	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 5 of 6 roles

systems

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 LOW Optional: Feature Engineering — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

Data Scientist

85.1 → 100.0 (+14.9 pts)

toolkit

presentation

different

problems

AI Engineer

81.1 → 100.0 (+18.9 pts)

models

technical

systems

teams

problems

detection

design

tools

Machine Learning Engineer

80.6 → 100.0 (+19.4 pts)

tools

design

models

applications

technical

systems

analysis

support

ML Engineer

72.3 → 98.4 (+26.1 pts)

systems

models

world

problems

technical

building

tools

detection

Applied Scientist

64.5 → 92.5 (+28.0 pts)

- computational
- research
- literature
- problems
- models
- technical
- systems
- design

MLOps Engineer

46.0 → 73.0 (+27.0 pts)

- detection
- systems
- preprocessing
- tools
- models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.